

- RAG Learning

2.5 Equations & Inequalities

Pearson Edexcel GCSE Mathematics 1MA1 — Algebra — spec refs A17–A22 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.5.1 Solving Linear Equations (A17)

An **equation** says that two expressions have the same value. It is only true for certain values of the unknown, and **solving** means finding those values. In a **linear** equation the highest power of x is 1, so there is exactly one solution (unless the equation is impossible or always true).

Balance method: keep the equation balanced by doing the same inverse operation to both sides, one step per line.

Two-step equations and brackets

Worked Example: Solve $6x - 7 = 29$.

Add 7 to both sides: $6x = 36$ Divide both sides by 6: $x = 6$

Worked Example: Solve $5(x + 4) = 45$.

Method 1 — expand first: $5x + 20 = 45 \Rightarrow 5x = 25 \Rightarrow x = 5$

Method 2 — divide first: $x + 4 = 9 \Rightarrow x = 5$

Unknown on both sides

Remove the *smaller* x term from both sides so that the x coefficient stays positive, then deal with the numbers.

Worked Example: Solve $8x - 3 = 5x + 18$.

Subtract $5x$: $3x - 3 = 18$

Add 3: $3x = 21$

$x = 7$ Check: $8(7) - 3 = 53$ and $5(7) + 18 = 53$ ✓

Equations with fractions

Multiply every term on both sides by the lowest common multiple of the denominators. Put each numerator in a bracket so that the multiplication reaches every term in it.

Worked Example: Solve $\frac{3x + 2}{4} = \frac{x + 6}{2}$.

Multiply both sides by 4: $3x + 2 = 2(x + 6)$

$3x + 2 = 2x + 12$

$x = 10$

Worked Example: Solve $\frac{x + 1}{4} + \frac{x + 2}{3} = 5$.

Multiply every term by 12: $3(x + 1) + 4(x + 2) = 60$

$3x + 3 + 4x + 8 = 60 \Rightarrow 7x + 11 = 60 \Rightarrow 7x = 49 \Rightarrow x = 7$

Approximate solutions from a graph

The equation $3x - 2 = 7$ is solved where the line $y = 3x - 2$ crosses the horizontal line $y = 7$, at $x = 3$. When the crossing point is not on a grid line, read the x -coordinate as accurately as the scale allows and say it is an **estimate**.

Exam Tip: Edexcel mark schemes give M1 for a correct first step (for example " $3x = 21$ ") even if the final answer is wrong, so write each step on a new line. On Paper 1 you have no calculator, so keep fractions exact rather than converting to rounded decimals.

2.5.2 Forming and Solving Equations (A21)

Problem-solving questions (often marked with P1 marks) expect you to **set up** the algebra yourself, **solve** it and then **interpret** the result in context.

1. Define the unknown, e.g. "let d be Dev's age now".
2. Write every quantity in terms of that unknown.
3. Use a known fact (angle sum, perimeter, total cost, equal amounts) to write an equation.
4. Solve it, then answer the question that was actually asked, with units.

Worked Example: The angles of a quadrilateral are x° , $3x^\circ$, $(2x + 15)^\circ$ and $(x + 30)^\circ$. Work out the size of the largest angle.

Angles in a quadrilateral add to 360° : $x + 3x + 2x + 15 + x + 30 = 360$

$$7x + 45 = 360 \Rightarrow 7x = 315 \Rightarrow x = 45$$

Angles: 45° , 135° , 105° , 75° . Largest angle = **135°** .

Worked Example: Carla is four times as old as Dev. In 6 years' time the sum of their ages will be 52. How old is Dev now?

Let Dev be d , so Carla is $4d$. In 6 years: $(d + 6) + (4d + 6) = 52$

$$5d + 12 = 52 \Rightarrow 5d = 40 \Rightarrow d = 8. \text{ Dev is } \mathbf{8} \text{ years old.}$$

Exam Tip: In age problems both people get older, so add the years to *each* person. Finish with a sentence that answers the question — " $x = 45$ " alone would not score the final mark for "the largest angle".

2.5.3 Quadratic Equations (A18)

A **quadratic equation** can be written in the form $ax^2 + bx + c = 0$ where $a \neq 0$. It has **two**, **one** (repeated) or **no** real solutions, which are the x -coordinates where the graph of $y = ax^2 + bx + c$ meets the x -axis.

Solving by factorising

If $A \times B = 0$ then $A = 0$ or $B = 0$. So with the equation equal to zero, factorise and set each factor equal to zero. At Foundation the equations are of the form $x^2 + bx + c = 0$ (including $x^2 + bx = 0$ and the difference of two squares).

Worked Example: Solve $x^2 - 11x + 24 = 0$.

Find two numbers with product $+24$ and sum -11 : -3 and -8 .

$$(x - 3)(x - 8) = 0 \Rightarrow x = 3 \text{ or } x = 8$$

Worked Example: Solve (a) $x^2 - 7x = 0$ (b) $x^2 - 36 = 0$.

(a) Common factor: $x(x - 7) = 0 \Rightarrow x = 0$ or $x = 7$ (never divide by x — that loses $x = 0$).

(b) Difference of two squares: $(x - 6)(x + 6) = 0 \Rightarrow x = 6$ or $x = -6$

Rearranging first, and $a \neq 1$ Higher

Worked Example: Solve $x^2 = 3x + 28$.

Make one side zero: $x^2 - 3x - 28 = 0$

$$(x - 7)(x + 4) = 0 \Rightarrow x = 7 \text{ or } x = -4$$

Worked Example: Solve $3x^2 + 10x - 8 = 0$.

$ac = 3 \times (-8) = -24$: two numbers with product -24 and sum 10 are 12 and -2 .

$$3x^2 + 12x - 2x - 8 = 3x(x + 4) - 2(x + 4) = (3x - 2)(x + 4) = 0$$

$$x = \frac{2}{3} \text{ or } x = -4$$

Completing the square

Higher

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

The number in the bracket is half the coefficient of x ; subtract its square to keep the expression equal.

Worked Example: Solve $x^2 + 8x + 3 = 0$, giving your answers in surd form.

$$(x + 4)^2 - 16 + 3 = 0 \Rightarrow (x + 4)^2 = 13$$

$$x + 4 = \pm\sqrt{13} \Rightarrow x = -4 \pm \sqrt{13}$$

If $a \neq 1$, factor a out of the x^2 and x terms first: $3x^2 + 12x - 5 = 3(x^2 + 4x) - 5 = 3[(x + 2)^2 - 4] - 5 = 3(x + 2)^2 - 17$.

The quadratic formula

Higher

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The **discriminant** $b^2 - 4ac$ tells you how many roots: > 0 two roots; $= 0$ one repeated root; < 0 no real roots. Learn the formula by heart.

Worked Example: Solve $3x^2 + 4x - 2 = 0$. Give your solutions correct to 2 decimal places.

$$a = 3, b = 4, c = -2$$

$$x = \frac{-4 \pm \sqrt{4^2 - 4(3)(-2)}}{2(3)} = \frac{-4 \pm \sqrt{16 + 24}}{6} = \frac{-4 \pm \sqrt{40}}{6}$$

$$x = \mathbf{0.39} \text{ or } x = \mathbf{-1.72} \text{ (2 d.p.)}$$

Exam Tip: "Give your answers correct to 2 decimal places" or "in the form $a \pm \sqrt{b}$ " is a signal that the quadratic will not factorise. Write the substitution out in full with negative values in brackets — that line alone earns the first M1.

Approximate solutions from a graph

To solve $x^2 - 4x + 1 = 0$ from the graph of $y = x^2 - 4x + 1$, read the x -values where the curve crosses the x -axis: approximately $x \approx 0.3$ and $x \approx 3.7$. To solve a different equation such as $x^2 - 4x + 1 = 2$ from the same graph, read where the curve meets the line $y = 2$ instead.

2.5.4 Simultaneous Equations (A19)

Simultaneous equations are a pair of equations in two unknowns that must both be satisfied at once. The solution is a pair of values, which is the **point of intersection** of their graphs.

Elimination method

Multiply one or both equations so that one variable has the same coefficient in each. If the signs are the same, subtract; if they are different, add.

Worked Example: Solve $3x + 2y = 12$ and $4x - 3y = 33$.

Multiply the first by 3: $9x + 6y = 36$

Multiply the second by 2: $8x - 6y = 66$

Add (the y terms have different signs): $17x = 102 \Rightarrow x = 6$

Substitute: $3(6) + 2y = 12 \Rightarrow 2y = -6 \Rightarrow y = -3$

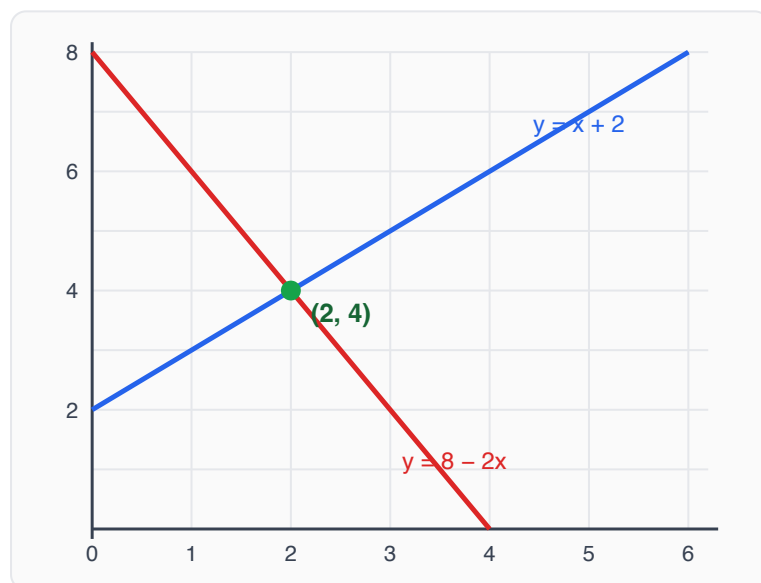
Check in the second equation: $4(6) - 3(-3) = 24 + 9 = 33 \checkmark$

Substitution method

Worked Example: Solve $y = 2x + 3$ and $3x + 2y = 20$.

Substitute for y : $3x + 2(2x + 3) = 20 \Rightarrow 7x + 6 = 20 \Rightarrow x = 2$, $y = 2(2) + 3 = 7$

Graphical solution



The lines $y = x + 2$ and $y = 8 - 2x$ intersect at $(2, 4)$, so the simultaneous equations have solution $x = 2$, $y = 4$. If the lines cross between grid lines, the graphical answer is only an approximation.

One linear and one quadratic equation Higher

Make y (or x) the subject of the **linear** equation and **substitute** it into the quadratic. Solve the resulting quadratic, then put each value back into the linear equation. A line can meet a curve at two points, so expect **two pairs** of answers.

Worked Example: Solve $y = x + 1$ and $y = x^2 - 2x - 3$.

$$x^2 - 2x - 3 = x + 1 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0$$

$$x = 4 \Rightarrow y = 5; \quad x = -1 \Rightarrow y = 0$$

Solutions: **(4, 5)** and **(-1, 0)**

Worked Example: Solve $x^2 + y^2 = 17$ and $y = x + 3$.

$$x^2 + (x + 3)^2 = 17 \Rightarrow 2x^2 + 6x + 9 = 17 \Rightarrow 2x^2 + 6x - 8 = 0 \Rightarrow x^2 + 3x - 4 = 0$$

$$(x + 4)(x - 1) = 0 \Rightarrow x = -4 \text{ or } x = 1$$

$$x = 1 \Rightarrow y = 4; \quad x = -4 \Rightarrow y = -1. \text{ Solutions: } \mathbf{(1, 4)} \text{ and } \mathbf{(-4, -1)}$$

Exam Tip: Present answers as matched pairs, e.g. " $x = 1, y = 4$ and $x = -4, y = -1$ ". Find the partner values from the *linear* equation; using $x^2 + y^2 = 17$ would give $y = \pm 4$ and introduce a wrong pair.

2.5.5 Iteration Higher (A20)

Iteration is a numerical method for finding an approximate solution. An equation is rearranged into the form $x = g(x)$ and used as an **iteration formula** $x_{n+1} = g(x_n)$.

Starting from a value x_1 (or x_0), each output is fed back in, and the values usually **converge** on a solution.

Showing a solution lies in an interval

If the value of the expression changes sign between $x = a$ and $x = b$ (for a continuous graph), then the equation "expression = 0" has a solution between a and b .

Worked Example: (a) Show that $x^3 - 5x - 3 = 0$ has a solution between $x = 2$ and $x = 3$.

$x = 2: 8 - 10 - 3 = -5$ $x = 3: 27 - 15 - 3 = 9$. There is a change of sign, so there is a solution between 2 and 3.

(b) Show that the equation can be rearranged to $x = \sqrt[3]{5x + 3}$.

$$x^3 = 5x + 3 \Rightarrow x = \sqrt[3]{5x + 3}$$

(c) Starting with $x_1 = 2$, use $x_{n+1} = \sqrt[3]{5x_n + 3}$ to find an estimate of the solution.

$$x_2 = \sqrt[3]{13} = 2.3513 \dots$$

$$x_3 = \sqrt[3]{5(2.3513 \dots) + 3} = 2.4528 \dots$$

$$x_4 = 2.4806 \dots, \quad x_5 = 2.4881 \dots, \quad x_6 = 2.4901 \dots \rightarrow \text{the solution is approximately } \mathbf{2.49}$$

Exam Tip: On your calculator: enter 2 and press =, then type the formula with ANS in place of x_n and press = repeatedly. Record each value to at least 4 decimal places, but never retype rounded values.

2.5.6 Linear Inequalities (A22)

An **inequality** uses $<$ (less than), $>$ (greater than), $\leq$ (less than or equal to) or $\geq$ (greater than or equal to). Its solution is a **range** of values rather than a single value.

Solving linear inequalities

Use the same steps as for an equation, with one extra rule: when you **multiply or divide by a negative number**, **reverse** the inequality sign. (Adding or subtracting never changes the sign.)

Worked Example: Solve $7 - 3x \leq 19$.

Subtract 7: $-3x \leq 12$

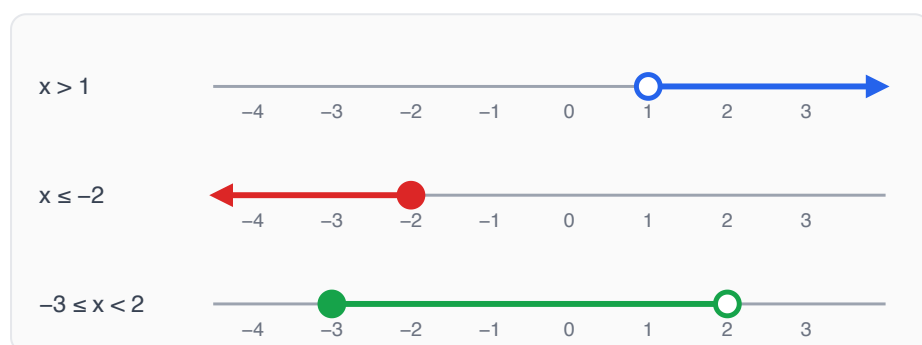
Divide by -3 and reverse the sign: $x \geq -4$

Worked Example: n is an integer such that $-9 \leq 3n < 6$. Write down all the possible values of n .

Divide every part by 3: $-3 \leq n < 2$

$n = -3, -2, -1, 0, 1$ (include -3 because of $\leq$; exclude 2 because of $<$)

Number lines



Symbol	Meaning	Number line
$<$ or $>$	strict — the end value is not included	open (empty) circle
$\leq$ or $\geq$	the end value is included	closed (filled) circle

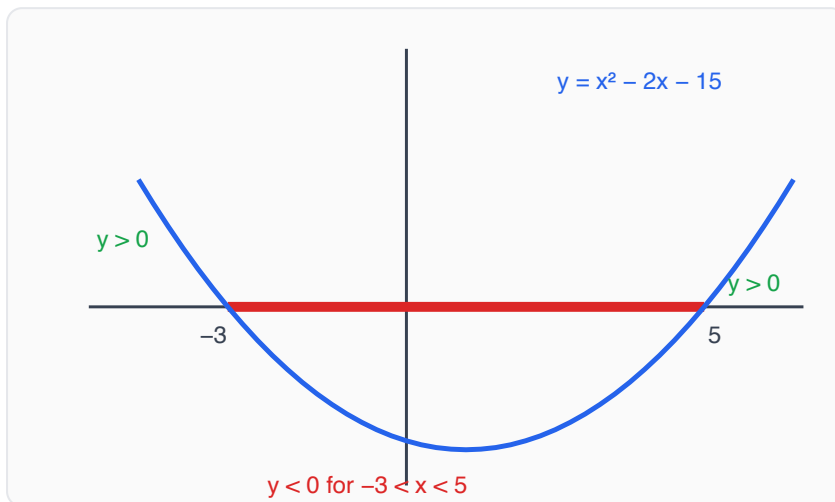
Set notation Higher

The 1MA1 Higher specification asks for solution sets in **set notation**. $\{x : x \geq -4\}$ is read "the set of all x such that x is greater than or equal to -4 ". A double inequality stays together: $\{x : -3 \leq x < 2\}$. Two separate parts are joined with the union symbol $\cup$: $\{x : x < -1\} \cup \{x : x > 6\}$.

2.5.7 Quadratic Inequalities Higher (A22)

1. Rearrange so that one side is 0, then solve the equation to find the **critical values**.
2. Sketch the graph (u-shaped when the x^2 coefficient is positive) through the critical values.

3. " < 0 " is where the curve is **below** the x -axis (between the critical values, one inequality); " > 0 " is where it is **above** (outside the critical values, two inequalities).



Worked Example: Solve $x^2 - 2x - 15 < 0$.

$(x - 5)(x + 3) = 0$ gives critical values $x = -3$ and $x = 5$.

The curve is below the x -axis between them: $-3 < x < 5$, or $\{x : -3 < x < 5\}$

Worked Example: Solve $3x^2 + 5x - 2 \geq 0$. Give your answer in set notation.

$(3x - 1)(x + 2) = 0 \Rightarrow x = \frac{1}{3}$ or $x = -2$

On or above the axis outside the critical values: $x \leq -2$ or $x \geq \frac{1}{3}$

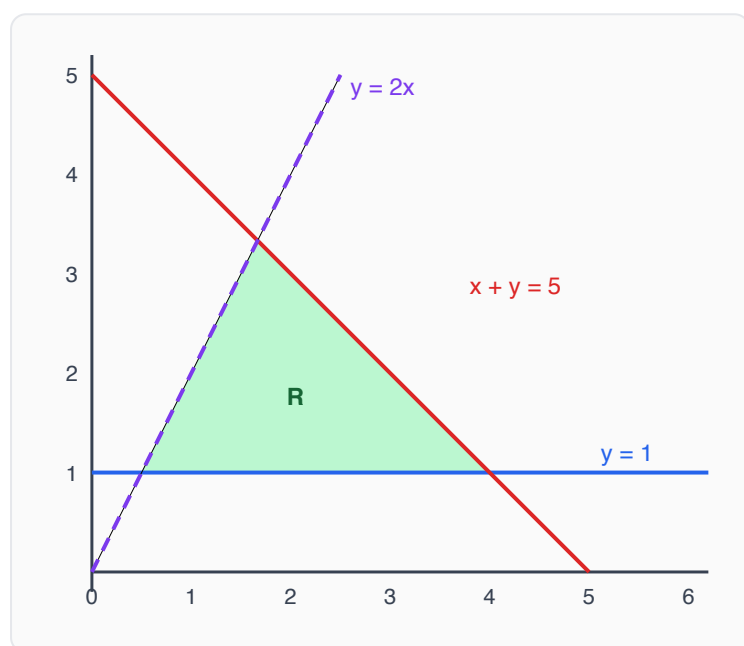
Set notation: $\{x : x \leq -2\} \cup \{x : x \geq \frac{1}{3}\}$

Exam Tip: " $-3 > x > 5$ " is a common wrong way to write an outside region — no number satisfies it. Use "or" (or $\cup$). Likewise $x^2 < 16$ means $-4 < x < 4$, not just $x < 4$.

2.5.8 Inequalities in Two Variables (Regions) Higher (A22)

An inequality such as $x + y \leq 5$ describes a **region** of the grid. Draw the boundary line $x + y = 5$, then test a point that is not on the line (the origin is usually easiest: $0 + 0 \leq 5$ is true, so the region is on the origin's side). Edexcel questions often ask you to shade and label the region R, or to list integer coordinates inside it.

Boundary lines: solid for $\leq$ or $\geq$ (points on the line are included); dashed (broken) for $<$ or $>$ (points on the line are not included).



Region R satisfies $y \geq 1$, $x + y \leq 5$ and $y < 2x$. The point $(2, 2)$ is in R: $2 \geq 1 \checkmark$, $2 + 2 = 4 \leq 5 \checkmark$, $2 < 4 \checkmark$. The point $(1, 2)$ is *not* in R because it lies on the dashed line $y = 2x$.

Exam Tip: For integer points, test every candidate in every inequality. Points on a solid line count; points on a dashed line do not.

Quick Reference Summary

Type	Method	Tier (1MA1)
Linear equation (A17)	Inverse operations; expand brackets; multiply through by the	Foundation

	LCM to clear fractions	
Forming equations (A21)	Define the unknown, form, solve, interpret in context	Foundation
Quadratic by factorising (A18)	Make $= 0$, factorise, set each factor to 0	Foundation ($x^2 + bx + c$) / Higher (rearrangement, $a \neq 1$)
Quadratic not factorising (A18)	Quadratic formula or completing the square	Higher
Linear simultaneous (A19)	Elimination, substitution or graph intersection	Foundation
Linear/quadratic simultaneous (A19)	Substitute the linear into the quadratic; two pairs of answers	Higher
Iteration (A20)	Sign change to locate a root; repeat $x_{n+1} = g(x_n)$	Higher
Linear inequality (A22)	As an equation; reverse the sign when $\times$ or $\div$ by a negative; number line	Foundation
Set notation, quadratic inequalities, regions (A22)	Critical values and a sketch; boundary lines and a test point	Higher